

SEVILLA AP, Spain

WMO: 083910

Lat: 37.4167N

Lon: 5.8792W

Elev: 111

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.0	38.8	22.2	16.8	50.2	26.9	21.0	51.9	23.1	56.4	20.5	56.9	3.2	40	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	26.9	102.6	70.6	100.1	70.0	97.2	69.2	75.6	93.1	73.7	90.0	72.3	87.9	7.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	116.3	80.2	69.6	109.2	78.4	68.0	103.1	77.1	39.0	92.8	37.3	89.3	36.0	86.9	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.4	15.4	31.9	108.4	3.1	3.3	29.6	110.8	27.8	112.7	26.1	114.6	23.8	117.0		
			29.9	76.7	3.6	2.9	27.3	78.8	25.2	80.5	23.1	82.2	20.5	84.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	67.2	52.1	54.3	59.4	64.0	70.6	77.8	82.7	83.3	77.0	70.2	59.5	54.2	(6)
(7)		DBStd	11.95	4.26	4.86	4.61	5.24	5.75	5.54	4.51	4.01	4.76	5.08	5.38	4.62	(7)
(8)		HDD50	58	25	15	1	0	0	0	0	0	0	0	2	15	(8)
(9)		HDD65	1502	399	300	180	80	17	0	0	0	0	14	177	335	(9)
(10)		CDD50	6327	91	136	293	419	638	834	1013	1034	810	627	288	144	(10)
(11)		CDD65	2296	0	0	7	49	190	384	548	569	360	176	13	0	(11)
(12)		CDH74	26019	0	3	129	536	2009	4449	6951	7088	3508	1293	53	0	(12)
(13)		CDH80	13830	0	0	13	134	852	2378	4149	4198	1673	430	3	0	(13)
(14)	Wind	WSAvg	6.1	5.4	6.3	6.6	6.8	6.7	6.7	6.5	6.1	5.7	5.2	5.4	5.7	(14)
(15)	Precipitation	PrecAvg	21.5	2.6	2.1	1.9	2.2	1.4	0.2	0.0	0.1	1.3	3.3	2.9	3.4	(15)
(16)		PrecMax	42.1	13.0	9.0	7.0	6.9	5.5	2.1	0.2	1.2	4.7	10.3	8.8	15.1	(16)
(17)		PrecMin	11.3	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	(17)
(18)		PrecStd	7.7	2.7	1.9	1.7	1.5	1.3	0.4	0.0	0.2	1.2	2.4	2.1	3.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	73.3	82.0	87.6	96.8	103.3	107.4	105.9	100.4	92.6	79.2	70.0	(19)
(20)		MCWB	55.8	57.2	60.5	63.0	66.5	70.3	74.9	72.2	69.9	64.6	63.9	58.3	(20)	
(21)		2%	DB	65.2	69.9	77.3	84.3	91.8	99.0	102.8	102.4	96.3	88.2	75.1	66.5	(21)
(22)		MCWB	55.2	56.5	59.0	61.7	64.7	69.0	70.7	70.5	68.8	64.7	62.2	58.5	(22)	
(23)		5%	DB	62.9	66.9	73.9	79.8	88.2	96.0	100.2	99.9	92.8	84.6	71.9	64.6	(23)
(24)		MCWB	54.9	55.5	57.7	61.0	63.9	68.1	70.3	70.0	67.5	64.6	60.7	57.8	(24)	
(25)		10%	DB	60.9	64.1	71.0	76.6	85.0	92.7	97.0	96.9	89.4	81.0	69.4	62.7	(25)
(26)		MCWB	54.1	54.6	56.8	60.0	63.1	67.4	69.4	69.5	67.1	64.6	60.1	56.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.5	62.6	63.2	67.2	71.3	76.2	79.3	79.0	74.2	72.2	69.7	64.4	(27)
(28)		MCDWB	63.3	65.0	75.1	78.6	86.5	95.3	102.2	96.8	88.7	76.5	73.9	65.8	(28)	
(29)		2%	WB	59.2	60.0	61.1	64.8	68.1	72.5	75.4	75.8	72.7	69.8	66.3	62.0	(29)
(30)		MCDWB	62.1	64.6	71.1	76.6	82.5	90.3	95.7	91.6	85.7	77.1	70.8	64.0	(30)	
(31)		5%	WB	57.1	58.5	59.8	63.1	66.2	70.6	73.3	73.8	71.3	68.2	63.5	59.7	(31)
(32)		MCDWB	60.6	63.1	69.0	74.4	82.0	87.9	92.8	88.7	83.2	75.6	68.5	62.6	(32)	
(33)		10%	WB	55.2	56.7	58.6	61.5	64.6	69.2	71.8	72.4	70.0	66.8	61.6	57.7	(33)
(34)		MCDWB	59.4	62.3	67.3	71.9	79.7	85.8	89.6	87.4	81.9	75.3	67.3	61.4	(34)	
(35)	Mean Daily Temperature Range		MDBR	17.9	19.3	20.8	21.2	23.4	25.4	27.6	26.9	22.8	20.0	17.9	16.8	(35)
(36)		5% DB	MCDBR	20.7	22.9	26.6	28.1	29.9	30.7	31.7	30.5	27.9	25.9	20.7	17.5	(36)
(37)			MCWBR	12.1	12.0	12.1	11.1	10.1	9.5	10.1	8.5	8.0	9.3	10.3	10.2	(37)
(38)			MCDBR	13.9	17.1	21.8	23.1	25.5	26.7	28.7	26.1	22.4	18.5	15.8	12.5	(38)
(39)		5% WB	MCWBR	10.0	10.4	11.1	10.4	10.4	10.1	11.4	10.0	8.3	8.9	9.8	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.322	0.344	0.382	0.381	0.395	0.405	0.385	0.401	0.395	0.373	0.340	0.326	(40)	
(41)		taud	2.483	2.427	2.320	2.344	2.332	2.322	2.361	2.327	2.366	2.438	2.490	2.491	(41)	
(42)		Ebn at Noon	274	281	279	285	282	278	282	276	271	267	265	264	(42)	
(43)		Edh at Noon	26	31	38	39	40	41	39	39	36	30	26	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	816	1080	1447	1797	2162	2450	2510	2206	1702	1212	886	726	(44)	
(45)		RadStd	98	139	151	132	162	103	48	67	105	103	85	77	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) -2.47	(i) -1.96	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	(47)

Nomenclature: See separate page